The observation scheme in the statistics of generalized repunit primes: calibrating the pooled test of the Lenstra–Pomerance–Wagstaff constant

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Abstract

The Lenstra–Pomerance–Wagstaff (LPW) heuristic predicts $\#\{n \le x : R_b(n) \text{ prime}\} \sim (e^\gamma/\ln b)\ln x$, where $R_b(n) = (b^n-1)/(b-1)$. Since 1993 this universality has been tested through a "normalized occurrence frequency" obtained by regressing $\ln n_k$ on k.

First, the suspicion that right-censoring biases this frequency downwards is *false*: both the endpoint form $D/(N\ln b)$ and the OLS slope are exactly unbiased (Proposition 2); only the reciprocal is biased, and that is Jensen's inequality rather than censoring. The genuine defect is different: competing forms of the estimator disagree by up to 29.5% on identical data—1.7 times the median deviation from $e^{-\gamma}$—and the "lucky base" label flips when the form is changed.

Second, an exact conditional likelihood replaces the regression and pools $B = 15$ bases into $M = 218$ events, against 51 for the Mersenne sequence alone. Our main result concerns the pivot $\kappa S \sim \text{Gamma}(M, 1)$: it is exact only under inverse sampling ("observe until the N-th event"), whereas the data arise from a fixed search frontier with a random number of finds. Under the actual scheme a nominal 5% test rejects in 16.2% of cases, and the correct pooled estimator is $(M-B)/S$, not $(M-1)/S$: the conditional likelihood costs one event *per base*. This yields $\hat\kappa = 1.838$ with a 95% interval [1.61, 2.13] and $p = 0.64$ against $e^\gamma \approx 1.781$; the reported ten-percent excess over LPW is an artifact of the observation scheme.

Third, the choice between the two estimators is settled by how the data were collected and not by the data themselves, and it costs 7%; hence our call to publish the search frontiers L_b. Finally, the homogeneity test ($\Lambda = 7.40$, $p = 0.92$) resolves only a spread of κ_b above ±34%, and the pooled sample only deviations from e^γ above 21%.

Keywords: repunit primes, generalized repunits, Mersenne primes, Lenstra–Pomerance–Wagstaff conjecture, Euler–Mascheroni constant, observation scheme, Type-I censoring, inverse sampling, Poisson point process, conditional likelihood, sufficient statistic, estimator bias, statistical power, computational number theory

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1 Introduction

For an integer base $b \ge 2$ the generalized repunit is

$$R_b(n) = \frac{b^n - 1}{b - 1}, \tag{1}$$

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and $R_b(n)$ can be prime only when n is prime. For $b = 2$ these are the Mersenne numbers $M_p = 2^p - 1$. Building on the heuristics of Wagstaff [1] and Pomerance [2], the LPW conjecture asserts that

$$\pi_b(x) = \#\{n \le x : \ R_b(n) \text{ prime}\} \ \sim \ \frac{e^\gamma}{\ln b} \ln x, \tag{2}$$

where γ is the Euler–Mascheroni constant; a detailed derivation of the heuristic for $b = 2$ is given in the PrimePages reference [3]. The base enters (2) only through the factor $1/\ln b$: the constant $e^\gamma \approx 1.7811$ is declared universal. Dubner [4] proposed testing this universality empirically across bases, and Bourdelais [5] has since maintained running estimates of the normalized frequency, publishing them in the comments to OEIS entries [6]. Related questions for the Mersenne sequence alone were discussed recently in [7]. Close in spirit is also [8], which looks for secondary structure in the Mersenne exponents related to the factorization of $p - 1$; methodologically it is built like our §8—stratified conditional likelihood plus a significance test—but it tests for structure *within* a single base, whereas we test for it *between* bases.

An interpretive folklore has grown up around these estimates. Base 13 is described as "lucky" and base 7 as "the unluckiest"; it is noted that the normalized frequencies lie below the conjectured limit $e^{-\gamma} \approx 0.5615$ and converge to it slowly from below. The present work examines these claims, but arrives at conclusions other than those one would naturally expect.

1.1 Contributions

1. **The estimator in common use is not biased.** The scheme in which the fitting interval ends at the largest known index does indeed condition the estimate on the last observation being an event, and the sufficient statistic is distributed as $\mathrm{Gamma}(N, \lambda_b)$ (Proposition 1). But this does *not* imply that the normalized frequency is biased: both the endpoint form $D/(N \ln b)$ and the OLS slope are exactly unbiased (Proposition 2, Table 2). Only the reciprocal $\hat\kappa = 1/\hat G$ is biased, by the factor $N/(N-1)$, and that is Jensen's inequality, not censoring. Unbiased estimators of κ and of $1/\kappa$ cannot be reciprocals of one another; any single "correction" that repairs one of them damages the other (§3).

2. **The genuine defect is instability with respect to the form of the estimator.** Several numerically distinct variants are in circulation. On identical data they disagree by as much as 29.5% (base 18)—1.7 times the median deviation of the estimates from $e^{-\gamma}$ (17.6%), which is what gets interpreted substantively. For two of the fifteen bases the choice of form weighs more than their entire deviation from the LPW prediction. Under the endpoint form base 18 is the third "unluckiest" of fifteen, and under OLS it is the tenth, below the median. The label describes the choice of estimator, not a property of the primes (§3.4).

3. **Pooling the bases, and what it costs.** The conditional likelihood (§3.5) does not require knowledge of the undocumented search bounds L_b and makes it possible to pool $B = 15$ bases into $M = 218$ events, against 51 for the Mersenne sequence alone. The exact null distribution $\mathrm{Gamma}(M, 1)$ makes simulation unnecessary.

4. **Main result: the pivot is not calibrated to the actual observation scheme.** The conditional likelihood is exact under the scheme "observe until the N-th event", where N is fixed by design. The data, however, arise from a fixed frontier: the search covered $(t_0, \ln L_b]$ and the number of finds is random. Under that scheme a nominal 5% test rejects in 16.2% of cases, and the correct pooled estimator is $(M - B)/S$ (Proposition 3). The distinction between the schemes is itself classical, and the unbiasedness of $(N-1)/D$ under scheme A is a textbook fact to which we lay no claim; what is new is that the frontier is *undocumented*, so the exposure must be reconstructed from the finds themselves, and the correction turns out to be proportional to the number of bases (Remark 3). This gives $\hat\kappa = 1.838$ with $p = 0.64$: the excess over LPW disappears (§4).

5. **The data do not settle the choice of scheme.** The estimators $(M-1)/S$ and $(M-B)/S$ are built on the same sufficient statistics (N_b, D_b) and differ by 7%; which of them is right is determined not by the data but by how the data were collected. An attempt to sidestep the choice by assembling the pooled estimate from per-base pieces that are unbiased by construction fails (§4.3). The only radical remedy is to publish the frontiers L_b.

6. **Lucky bases: what the test cannot do.** The likelihood-ratio test gives $\Lambda = 7.40$ on 14 degrees of freedom, $p = 0.92$. But its power is such that it resolves only a spread of κ_b of order $\pm 34\%$ (§5.3). The correct conclusion is "base-dependent frequencies are not required by the data", not "they do not exist".

7. **An explicit power calculation.** The pooled sample resolves deviations from e^γ of at least 21.2%; the Mersenne sequence alone, at least 49.8% (§6).

Section 7 works through a narrower but instructive episode: a naive regression of the Mersenne counting function produces an apparently significant negative intercept, which turns out to be an artifact of conditioning. We include it as a worked example of the same error in a different guise.

2 Model and data

2.1 The Poisson model

Put $t = \ln n$. The heuristic (2) is equivalent to the statement that the indices of generalized repunit primes to base b form a homogeneous Poisson point process on the t-axis with constant intensity

$$\lambda_b = \frac{\kappa}{\ln b}, \qquad \text{LPW:} \;\; \kappa = e^\gamma = 1.781072\ldots \tag{3}$$

Universality is the assertion that a single parameter κ governs all bases. The Poisson idealization here is a working assumption, not a theorem: even for the ordinary primes the Poisson law in short intervals has been established only conditionally, under the Hardy–Littlewood conjectures, and it is known to break down as the scale grows [9]. We adopt (3) as a model and treat κ throughout as a quantity to be estimated; this reduces the question "is LPW true?" to a one-parameter one and makes the bases poolable.

2.2 Admissible bases

Bases that are perfect powers are excluded: for $b = c^m$ with $m > 1$ the number $R_b(n)$ admits an algebraic factorization and the process does not have the assumed form. Bases $4, 8, 9, 16$ are therefore omitted. Within the declared range the selection does not depend on the results: we take *all* bases $2 \le b \le 20$ that are not perfect powers, which gives $B = 15$; no base was excluded or added on the strength of its estimate. Each of them has at least eight known indices (the minimum is attained at $b = 18$: eight terms of A133857).

We stress, however, that the upper limit $b \le 20$ is itself a conventional round number and not a consequence of the criterion "at least eight indices": that criterion continues to hold beyond it (A127996 for $b = 21$ has eight terms, A127997 for $b = 22$ has nine) and first fails at $b = 23$ (A204940, six terms). To make sure the limit does not look chosen after the fact, we tested its influence directly. Extending the set to all bases $b \le 26$ that are not perfect powers (this adds $b = 21, 22, 23, 24, 26$; $b = 25 = 5^2$ is excluded algebraically) gives $B = 20$, $M = 253$ and

$$\frac{M-1}{S} = 1.9776, \qquad \frac{M-B}{S} = 1.8285,$$

that is, it shifts the main estimate (15) by 0.5%—more than an order of magnitude less than the price of the choice of observation scheme (§4). The homogeneity test on the extended set gives $\Lambda = 10.27$ on 19 degrees of freedom, $p = 0.95$. No conclusion of this work depends on the limit $b \leq 20$; we retain it as pre-declared.

2.3 The data and their status

Table 1 lists the sequences used, retrieved on 27 August 2026 (cross-check: Appendix A). Three caveats matter.

Probable primes. For large indices most of the entries are probable primes established by a Fermat or strong Fermat test after trial division. The count is biased upwards only if a Fermat pseudoprime has been mistaken for a prime, which for numbers of this size is practically unheard of. Sensitivity is examined in §5.4.

Verification frontiers. For $b = 2$ the GIMPS project [10] publishes two frontiers: all exponents below $X_{\text{ver}} = 81\,648\,221$ have been tested twice, which gives exactly 50 Mersenne primes, whereas all exponents below $X_{\text{ft}} = 141\,308\,443$ have been tested at least once, which gives 52. The two largest exponents are provisional: unverified candidates remain between them. The analysis of §7, where the frontier is used explicitly, is restricted to the 50 verified exponents. The pooled analysis is insensitive to this choice (§5.4).

Both frontiers are moving quantities, so we fix the moment of reading: the PrimeNet Milestones report, 27 August 2026. No conclusion depends on the exact value of X_{ver}: it suffices that it lies between $M_{50} = 77\,232\,917$ and $M_{51} = 82\,589\,933$, and this holds for every value the report has taken during 2026.

The status of entry A000043. The DATA section of that entry contains 50 terms and stops at 77 232 917: OEIS regards the order of the subsequent exponents as unproven until double-checking is complete. The exponents 82 589 933 and 136 279 841 are known and documented by GIMPS (and are mentioned in the comments to A000043), but formally they are not part of DATA. We include them, because the statistic (8) uses only the position of the last event and not its ordinal number; §5.4 verifies that restricting base 2 to the fifty double-checked exponents does not change the conclusion.

Search bounds for $b > 2$. For the other bases no comparable frontier is published. It is precisely this gap that is the subject of §4, and, we believe, the principal obstacle to a quantitative test of LPW.

3 What is biased and what is not

3.1 The estimators in common use

Dubner [4] and Bourdelais [5] characterize a base b by a normalized occurrence frequency obtained by fitting a straight line to $\ln n_k$ against k and dividing by $\ln b$. By (3) the expected slope is $1/\lambda_b = \ln b/\kappa$, so the normalized frequency estimates

$$G_b = \frac{1}{\kappa}, \qquad \text{LPW:}\;\; G_b = e^{-\gamma} = 0.5614595\ldots \tag{4}$$

Several variants are in circulation—ordinary OLS with an intercept, regression through the origin, and the endpoint form $\hat{G}_b = D_b/(N_b \ln b)$ with $D_b = t_{N_b} - t_0$. All of them share one structural feature: the fitting interval ends at the largest known index.

Table 1. Sequences used. N_b is the number of events strictly to the right of $t_0 = \ln n_{\min}$, that is, one less than the number of known prime indices: the smallest index sets the origin, not an event. In the last row $M = \sum_b N_b$ and $S = \sum_b D_b / \ln b$; the quantity S is not the sum of the D_b column. Source: OEIS entries, retrieved 27 August 2026.

b	OEIS	primes	N_b	$n_{\min}$	$n_{\max}$	D_b
2	A000043	52	51	2	136 279 841	18.0371
3	A028491	23	22	3	8 530 117	14.8605
5	A004061	19	18	3	3 300 593	13.9110
6	A004062	17	16	2	3 360 347	14.3344
7	A004063	10	9	5	1 264 699	12.4409
10	A004023	11	10	2	8 177 207	15.2237
11	A005808	13	12	17	1 868 983	11.6077
12	A004064	14	13	2	769 543	12.8604
13	A016054	13	12	5	1 503 503	12.6139
14	A006032	11	10	3	1 724 417	13.2618
15	A006033	10	9	3	639 833	12.2704
17	A006034	12	11	3	1 990 523	13.4053
18	A133857	8	7	2	1 270 141	13.3615
19	A006035	11	10	19	209 359	9.3074
20	A127995	9	8	3	984 349	12.7011
total			$M = 218$			$S = 110.4423$

3.2 The exact distribution

Proposition 1 (Conditional law of the right endpoint). *Let the events form a homogeneous Poisson process of intensity λ on (t_0, ∞), and let $t_1 < \cdots < t_N$ be its first N events after t_0. Then $D := t_N - t_0$ has the* Gamma(N, λ) *distribution with density $\lambda^N d^{N-1} e^{-\lambda d}/(N-1)!$. Conditionally on $D = d$, the preceding $N-1$ events are distributed as the order statistics of $N-1$ independent uniform variables on (t_0, d).*

Proof. The waiting times $t_1 - t_0, \ldots, t_N - t_{N-1}$ are independent Exp(λ) variables, and a sum of N of them is Gamma(N, λ). Conditional uniformity is the order-statistics property of the Poisson process, applied to (t_0, d). □

Corollary 1 (Bias of the reciprocal). *The maximum-likelihood estimator of λ based on D alone is $\hat{\lambda} = N/D$, and*

$$\mathbb{E}[\hat{\lambda}] = N\,\mathbb{E}[D^{-1}] = \frac{N}{N-1}\,\lambda, \tag{5}$$

since $\mathbb{E}[D^{-1}] = \lambda/(N-1)$ for $D \sim$ Gamma(N, λ). The estimator $\tilde{\lambda} = (N-1)/D$ is exactly unbiased.

3.3 What does not follow from Corollary 1

It is natural to conclude that, since $\hat{\lambda}$ is inflated by the factor $N/(N-1)$, the frequency $\hat{G}_b$ must be deflated by the same factor. This is false.

Proposition 2 (Unbiasedness of the frequency). *Under the hypotheses of Proposition 1:*

(i) *the endpoint estimator is exactly unbiased:* $\mathbb{E}\left[\dfrac{D}{N \ln b}\right] = \dfrac{1}{\lambda \ln b} = G_b$;

(ii) *the OLS slope in the regression of t_k on k is exactly unbiased:* $\mathbb{E}[\hat{\beta}] = 1/\lambda$;

Table 2. Monte Carlo check of the propositions, 200 000 replications per row (generator seed fixed, see `analysis.py`). The target value of every column except the second is 1.0000.

N	$\mathbb{E}[\hat{\lambda}]/\lambda$	$N/(N-1)$	$\mathbb{E}[\tilde{\lambda}]/\lambda$	$\mathbb{E}[\hat{G}]\lambda$	$\mathbb{E}[\tilde{G}]\lambda$	$\mathbb{E}[\hat{\beta}]\lambda$
7	1.1665	1.1667	0.9998	1.0002	1.1669	1.0002
10	1.1115	1.1111	1.0004	0.9998	1.1109	0.9997
16	1.0672	1.0667	1.0005	0.9994	1.0660	0.9994
22	1.0481	1.0476	1.0004	0.9996	1.0472	0.9997
51	1.0202	1.0200	1.0002	0.9999	1.0199	0.9998

(iii) *consequently the "corrected" quantity $\tilde{G}_b = \frac{N}{N-1}\hat{G}_b$ is biased* upwards*, by exactly the factor $N/(N-1)$.*

Proof. (i) $\mathbb{E}[D] = N/\lambda$ for $D \sim \mathrm{Gamma}(N, \lambda)$, whence the result at once. (ii) The OLS slope is the linear statistic $\hat{\beta} = \sum_k w_k t_k$ with weights $w_k = (k - \bar{k})/\sum_j (j - \bar{k})^2$, for which $\sum_k w_k = 0$ and $\sum_k w_k k = 1$. Since $\mathbb{E}[t_k] = t_0 + k/\lambda$, we get $\mathbb{E}[\hat{\beta}] = \sum_k w_k\,(t_0 + k/\lambda) = 1/\lambda$. Part (iii) is immediate from (i). □

The substantive point is simple: unbiased estimators of κ and of $1/\kappa$ cannot be reciprocals of one another, because $x \mapsto 1/x$ is strictly convex. The unbiased pair is $(\hat{G}_b,\ \tilde{\kappa}_b)$ with $\tilde{\kappa}_b = (N_b - 1)\ln b/D_b$; but $\tilde{\kappa}_b \neq 1/\hat{G}_b$. Any single "censoring correction" that repairs one of these quantities damages the other. Table 2 confirms Proposition 2 by simulation.

Remark 1. The negative result of this subsection deserves emphasis, because it runs against intuition: the systematic downward deviation of the published frequencies from $e^{-\gamma}$ is *not* explained by the fitting scheme. Its source is either a genuine excess of κ over e^{γ} or, as §4 shows, the observation scheme—but not the arithmetic of the estimator.

3.4 Instability with respect to the form of the estimator

Unbiasedness does not mean agreement. Table 5 gives the endpoint form $\hat{G}_b$ and the OLS estimate $\hat{G}_b^{\mathrm{OLS}}$ on identical data. The discrepancies reach 29.5% ($b = 18$), 15.9% ($b = 13$) and 10.3% ($b = 11$).

Compare these with what the literature interprets substantively—the deviations of the estimates themselves from $e^{-\gamma}$. Across the fifteen bases the latter have median 17.6% (from 5.6% at $b = 20$ to 43.7% at $b = 19$). The largest discrepancy between forms, 29.5%, exceeds that median by a factor of 1.7. On average the choice of form is of course cheaper: the median discrepancy is 4.1%, about a quarter of the median deviation. But for two of the fifteen bases ($b = 18$ and $b = 20$) the choice of form weighs more than their entire deviation from the LPW prediction—and one of them, base 18, is precisely a base the folklore discusses.

The consequence for the folklore is direct. Under the endpoint form base 18 has $\hat{G}_{18} = 0.6604$, the third largest of fifteen (after $b = 7$ with 0.7104 and $b = 10$ with 0.6612), so it belongs firmly among the three "unluckiest"; under OLS, $\hat{G}_{18}^{\mathrm{OLS}} = 0.4656$, tenth of fifteen and below the median 0.4904, so it has become "lucky". The label flips without a single new prime, purely from a change in the form of the estimator: the base moves from third place to tenth.

To this is added a second, technical difficulty with OLS: its standard errors are wrong in this problem. The residuals are deviations of order statistics and are strongly autocorrelated; for $b = 2$ we obtain $\rho_1 = 0.825$, and the Newey–West covariance estimator with eight lags raises the standard error of the intercept from 0.447 to 0.732 (§7). Published point values come with no intervals at all, and this—not bias—is the principal methodological objection to current practice.

3.5 The conditional likelihood

Proposition 1 yields more than an analysis of bias. Setting $t_0 = \ln n_{\min}$ and taking (N_b, D_b) as the data for base b, we obtain the log-likelihood

$$\ell(\kappa) = \sum_b \Big[N_b \ln \frac{\kappa}{\ln b} - \frac{\kappa}{\ln b} D_b \Big] + \text{const}, \tag{6}$$

whose distribution does not depend on the search bound L_b. This matters in practice: L_b is well documented only for $b = 2$. Differentiating (6) and putting $S = \sum_b D_b / \ln b$, we get

$$\hat{\kappa} = \frac{M}{S}, \qquad \tilde{\kappa} = \frac{M-1}{S}, \qquad M = \sum_b N_b. \tag{7}$$

Since $\kappa S = \sum_b \lambda_b D_b$ is a sum of independent Gamma$(N_b, 1)$ variables,

$$\kappa S \sim \text{Gamma}(M, 1), \tag{8}$$

an exact pivot; confidence intervals and p-values follow in closed form. *It is precisely this statement that requires revision*—see §4.

Remark 2 (Independence across bases). Pooling presupposes that the processes for different bases are independent. Strictly this fails at small indices: seventeen index values occur in more than one base, and $n = 3$ occurs in nine of them, since $R_b(3) = b^2 + b + 1$ is prime for many b. All such coincidences lie at $n \leq 317$. More important than coinciding indices (the numbers $R_b(n)$ for different b are distinct) is the possible sharing of algebraic divisors; both affect only small n, whereas D_b is determined by the upper end of the range. Section 5.4 confirms that discarding all indices below 10^3 leaves the conclusion unchanged.

4 Two observation schemes

4.1 The distinction

The pivot (8) was derived on the assumption that N_b is fixed by the observation plan: we watch the process until N_b events have accumulated, then stop. Call this *scheme A* (inverse sampling, Type-II censoring).

The data arose differently. For each base the computational search covered the whole interval $(t_0, \ln L_b]$, and every event falling inside it was found; the number of finds N_b is a random variable, while the frontier L_b is fixed. Call this *scheme B*.

Remark 3 (What here is classical and what is not). The distinction between the schemes is itself classical and goes back to reliability theory, where scheme A is called a *failure-truncated* and scheme B a *time-truncated* life test [11, 12]. The estimator $(N-1)/D$, unbiased under scheme A (Corollary 1), is a textbook fact there, and we lay no claim to it. It is also known that under scheme B no exactly unbiased estimator of the *mean* $1/\lambda$ exists at all [13, 14]; that is a warning about how delicate the question of unbiasedness becomes under Type-I censoring. We stress, however, that the remainder in (11) has nothing to do with that theorem: by Remark 4 it is the expected number of bases yielding no find at all, and its origin is quite different.

What we regard as new is something else. In the classical setting the moment at which the test is stopped is *known*: the time T is set by the experimenter. Here the frontier L_b is undocumented for every base except $b = 2$, so the full exposure cannot be read off a protocol and must instead be reconstructed—via the memorylessness of the process—from the finds themselves. That is where a correction *per base*, rather than one for the sample as a whole, comes from: there are as many unknown frontiers as there are bases. A survey of the biases generated by the edge of the observation window, though in a nonparametric setting and for the shape of the interevent distribution rather than for the intensity, is given in [15].

The distinction is not cosmetic. Under scheme B, conditionally on $N_b = n$, the positions of the events are n independent uniform variables on $(t_0, \ln L_b]$, so the conditional distribution of D_b depends on L_b and *not* on λ_b: with N_b fixed, D_b is informative about the frontier, not about the intensity. Under scheme B the likelihood (6) is not a likelihood.

Note also that §7 of the present work, where the frontier for $b = 2$ is known and used explicitly, rests on scheme B. One cannot apply both schemes in a single paper; below we adopt scheme B as matching the way the data were obtained, and calibrate the scheme A results to it.

4.2 The correct pooled estimator under scheme B

The bias has a closed form, and an instructive one.

Proposition 3. *Suppose that under scheme B the search for base b covered $(t_0, \ln L_b]$, let $W_b = \ln L_b - t_0$ be the width of the window, and let $\delta_b = \ln L_b - t_{N_b}$ be the distance from the frontier to the last event found (with $\delta_b = W_b$ when $N_b = 0$). Then*

$$\Pr(\delta_b > x) = e^{-\lambda_b x} \quad (0 \le x < W_b), \qquad \Pr(\delta_b = W_b) = e^{-\lambda_b W_b}, \tag{9}$$

that is, δ_b is an exponential variable with parameter λ_b truncated at W_b, and

$$\mathbb{E}[\delta_b] = \frac{1 - e^{-\lambda_b W_b}}{\lambda_b}. \tag{10}$$

The full exposure is $S_{\text{full}} = S + \sum_b \delta_b / \ln b = \sum_b W_b / \ln b$, with $\mathbb{E}[M] = \kappa S_{\text{full}}$ identically, and

$$\mathbb{E}\Big[\kappa \sum_b \delta_b / \ln b\Big] = B - \sum_b e^{-\lambda_b W_b}. \tag{11}$$

Equating κS_{full} to M and dropping the exponentially small remainder gives the method-of-moments estimator

$$\hat{\kappa}_{\text{B}} = \frac{M - B}{S}. \tag{12}$$

Proof. For $x < W_b$ the event $\{\delta_b > x\}$ means that the interval $(\ln L_b - x, \ln L_b]$ of length x contains no event of the process, which has probability $e^{-\lambda_b x}$. As $x \to W_b^-$ what remains is the case "no events at all", which gives the atom in (9). Integrating the tail, $\mathbb{E}[\delta_b] = \int_0^{W_b} e^{-\lambda_b x}\, dx = (1 - e^{-\lambda_b W_b})/\lambda_b$, which is (10). Next, S_{full} is not random: by construction $D_b + \delta_b = W_b$, so $\kappa S_{\text{full}} = \sum_b \lambda_b W_b = \mathbb{E}[M]$—this is exactly the moment equation. Finally $\kappa\, \mathbb{E}[\delta_b]/\ln b = \lambda_b \mathbb{E}[\delta_b] = 1 - e^{-\lambda_b W_b}$, and summation over b gives (11). □

Remark 4. The remainder $\sum_b e^{-\lambda_b W_b}$ dropped in (11) is the expected number of bases for which the search produced no find at all. In our data every base has finds and $\lambda_b W_b \approx N_b \ge 7$, so the remainder does not exceed $\sum_b e^{-N_b} < 3 \cdot 10^{-3}$ against $B = 15$—a relative correction of order 0.02%. That is what the first row of Table 3 exhibits. We stress that *without* the truncation the statement $\delta_b \sim \text{Exp}(\lambda_b)$ would be false: the process has no events to the left of t_0, so $\delta_b \le W_b$ by construction, and an appeal to memorylessness alone is not enough here.

In other words, the conditional likelihood costs information equivalent to one event *per base*—whereas formula (7) subtracts one event *in total*. With $B = 15$ the difference is fourteen events, that is, 7%:

$$\frac{M-1}{S} = 1.9648 \qquad \text{versus} \qquad \frac{M-B}{S} = 1.8381. \tag{13}$$

Direct simulation of scheme B confirms this: estimator (12) has a bias of +0.02%, whereas $(M-1)/S$ has a bias of +6.9% (Table 4).

It is essential that the conclusion of Proposition 3 rests *only* on identity (10), which holds for *any* position of the frontier: only the window width W_b enters it, and that cancels together

Table 3. Unbiasedness of the two pooled estimators under different frontier placements. True value $e^{\gamma} = 1.7811$; 15 000 replications per row. The rules are stated through the window width $W_b = \ln L_b - t_0$: "$\mathbb{E}[\#]$ = observed" is $W_b = N_b/\lambda_b$; "$+s$ mean intervals" is $W_b = (N_b + s)/\lambda_b$; "$c$ times further in t" is $W_b = c\,D_b$ with the observed D_b; "random" is $W_b = (N_b/\lambda_b)\,U$ with independent $U \sim \mathrm{Unif}(0.7, 1.8)$, drawn separately for each base and each replication.

Rule for the frontier	$\mathbb{E}[(M-B)/S]$	dev.	$\mathbb{E}[(M-1)/S]$	dev.
$\mathbb{E}$[#events] = observed	1.7814	+0.02%	1.9043	+6.92%
frontier +0.5 mean intervals	1.7816	+0.03%	1.9001	+6.68%
frontier +2 mean intervals	1.7820	+0.05%	1.8891	+6.07%
frontier +5 mean intervals	1.7820	+0.05%	1.8717	+5.09%
frontier 1.5 times further in t	1.7825	+0.08%	1.8716	+5.08%
frontier twice as far in t	1.7816	+0.03%	1.8475	+3.73%
random frontiers, differing by base	1.7812	+0.01%	1.8788	+5.49%

with the exponentially small remainder (Remark 4). Estimator (12) therefore contains no free parameter and does not require knowing how far the frontier has advanced. Table 3 confirms this by simulation under seven different rules for placing the frontier, including random ones and ones differing between bases.

The only assumption. All that is required is the following.

(A1) *The position of the frontier L_b was not chosen in the light of the search results: the computational budget, not the finds, determines how far the sieving has been carried.*

In the language of survival analysis, (A1) is the requirement that the censoring be *non-informative*: the stopping time does not depend on what the process has produced. A violation of (A1) is exactly informative censoring and, as usual in such problems, biases the estimate in the direction dictated by the stopping rule; the size of the bias is written out below.

If (A1) fails in the extreme form—the search for each base being stopped immediately after a find—then $\delta_b = 0$ and we are back at scheme A. The general form of the estimator when $\delta_b = m/\lambda_b$ is $\hat{\kappa} = (M - Bm)/S$, so the whole range generated by the uncertainty in (A1) is the segment between 1.838 ($m = 1$, frontier independent of the finds) and 1.974 ($m = 0$, search stopped at a find).[1] Actual practice in distributed computing is clearly closer to the first: sieving proceeds over contiguous intervals and finds are announced as they occur. It is essential that both ends of the segment are compatible with LPW (§5.2), so the conclusion of this work does not depend on (A1); what depends on (A1) is only the size of the residual gap.

Two conclusions follow from Table 4. First, under scheme B a nominal 5% test rejects a true hypothesis in 16.2% of cases: the claimed precision of the intervals is overstated roughly threefold. Second, under scheme B estimator (12) is practically unbiased, so the issue is not an irremovable distortion but a wrongly chosen scheme.

The only direct check available is for $b = 2$: there $\delta_2 = \ln(X_{\mathrm{ver}}/M_{50}) = 0.056$ against the mean interval $\ln 2/e^{\gamma} = 0.389$ predicted by LPW. This is a single realization of an exponential variable with unit mean, and such a value is not unusual: $\Pr(\delta\lambda < 0.143) = 0.13$. (Taking the estimated rather than the predicted mean interval, $1/\hat{\lambda} = 0.358$, gives $\delta\lambda = 0.155$ and $\Pr = 0.14$; the conclusion is the same.) Note that the unbiasedness of estimator (12) does not depend on this realization—it averages over all fifteen bases—and that the variability of δ_b itself is accounted for in the confidence interval (15), obtained by inverting a simulation of scheme B.

[1] For $m = 0$ the moment equation gives exactly $M/S = 1.9739$, whereas the exactly unbiased scheme A estimator of Corollary 1 is $(M-1)/S = 1.9648$ (the difference is one event out of 218, that is, 0.5%; this is the usual gap between a moment estimator and one corrected for Jensen's inequality). In §5.2 and everywhere below we report $(M-1)/S$; here, for homogeneity within the family $(M - Bm)/S$, the limit of the moment form is written.

Table 4. Calibration of the pivot (8) under scheme B. "Frontier shift" is how many mean intervals beyond the expected last event $\ln L_b$ is placed. 20 000 replications per row; Monte Carlo accuracy about ± 0.005 (binomial at the observed level 0.16; the spread across five generator seeds is 0.162–0.169).

Scheme	type I error (nominal 0.05)	$\mathbb{E}[\tilde{\kappa}]/\kappa = c$
B, shift 0	0.1616	1.0688
B, shift 0.5	0.1600	1.0664
B, shift 1	0.1558	1.0641
B, shift 2	0.1491	1.0595
A (control)	0.0504	0.9997

4.3 The data do not choose the scheme for us

We stress that $(M-1)/S$ and $(M-B)/S$ are functions of the same sufficient statistics (N_b, D_b). No processing of the available data will say which of them is right: the choice is determined by how the data were collected, and its price is 7%—twice the entire residual gap with LPW.

The natural attempt to sidestep the choice is to build the pooled estimate out of pieces that are unbiased by construction. The per-base estimator $\tilde{\kappa}_b = (N_b - 1)\ln b/D_b$ is unbiased under scheme A (Corollary 1) and unbiased under scheme B up to an exponentially small term. Indeed, for $N_b = n \geq 2$ the ratio D_b/W_b is distributed as $\mathrm{Beta}(n, 1)$, and the identity $\mathbb{E}[1/\mathrm{Beta}(\alpha, \beta)] = (\alpha + \beta - 1)/(\alpha - 1)$ gives $\mathbb{E}[\tilde{\lambda}_b \mid N_b = n] = n/W_b$; for $n \in \{0, 1\}$ the estimator is zero. Averaging over $n \sim \mathrm{Pois}(\lambda_b W_b)$ therefore gives not λ_b but

$$\mathbb{E}[\tilde{\lambda}_b] = \lambda_b - \frac{\Pr(N_b = 1)}{W_b} = \lambda_b\Big(1 - e^{-\lambda_b W_b}\Big),$$

the same exponentially small deficit as in (11): for $\lambda_b W_b \approx N_b \geq 7$ it does not exceed 0.1%. Simulation confirms this: for each of the fifteen bases the mean of $\tilde{\kappa}_b$ lies within 1.774–1.790 against the true value 1.781 (the estimator has infinite variance when $N_b = 2$, so the sample mean converges slowly; the residual spread across bases is Monte Carlo noise).

Nevertheless, the weighted mean

$$\kappa_* = \sum\nolimits_b (N_b - 2)\tilde{\kappa}_b \Big/ \sum\nolimits_b (N_b - 2)$$

with inverse-variance weights has a bias of $+0.14\%$ under scheme A but $+7.51\%$ under scheme B, that is, essentially the same as $(M-1)/S$. The reason is that the weights—the event counts themselves—are random under scheme B and positively correlated with the estimates; unbiasedness of each summand does not carry over to a ratio of sums.

From this we draw a narrow but, we think, important conclusion: *the gain in precision from pooling the bases is inseparable from an assumption about the observation scheme.* The only radical remedy is to document the search frontiers L_b, as GIMPS does for $b = 2$. We address this request to the participants of the relevant search projects.

5 Results

5.1 Per-base estimates

Eleven of the fifteen bases lie below $e^{-\gamma}$—the same eleven under either form of the estimator, so the sign of the deviation is robust to the choice of form, unlike the ordering of the bases among themselves. All fifteen PIT quantiles lie inside the central 95% region; the extremes are $b = 19$ at quantile 0.061 and $b = 7$ at 0.801. Base 13, the "lucky" one, sits at 0.174, which is in no way remarkable for one base out of fifteen.

Table 5. Per-base estimates. $\hat{G}_b = D_b/(N_b \ln b)$ is the endpoint form; $\hat{G}_b^{\text{OLS}}$ is the OLS slope of $\ln n_k$ on k divided by $\ln b$; both are unbiased (Proposition 2). $\tilde{\kappa}_b = (N_b - 1)\ln b/D_b$ is the unbiased estimator of the constant, with an exact interval from Gamma$(N_b, 1)$. The last column is the probability integral transform of Gamma$(N_b, 1)$ applied to $e^{\gamma} D_b/\ln b$. LPW predicts $G = e^{-\gamma} = 0.5615$ and $\kappa = e^{\gamma} = 1.7811$.

b	N_b	$\hat{G}_b$	$\hat{G}_b^{\text{OLS}}$	diff., %	$\hat{\kappa}_b$	$\tilde{\kappa}_b$	95% CI	PIT
2	51	0.5102	0.5309	+4.1	1.960	1.921	[1.459, 2.533]	0.266
3	22	0.6148	0.6081	−1.1	1.626	1.552	[1.019, 2.373]	0.693
5	18	0.4802	0.4666	−2.8	2.083	1.967	[1.234, 3.149]	0.285
6	16	0.5000	0.4904	−1.9	2.000	1.875	[1.143, 3.092]	0.355
7	9	0.7104	0.7371	+3.8	1.408	1.251	[0.644, 2.466]	0.801
10	10	0.6612	0.6746	+2.0	1.512	1.361	[0.725, 2.584]	0.737
11	12	0.4034	0.3617	−10.3	2.479	2.272	[1.281, 4.066]	0.162
12	13	0.3981	0.3796	−4.7	2.512	2.319	[1.337, 4.050]	0.141
13	12	0.4098	0.3448	−15.9	2.440	2.237	[1.261, 4.002]	0.174
14	10	0.5025	0.5386	+7.2	1.990	1.791	[0.954, 3.400]	0.406
15	9	0.5035	0.5054	+0.4	1.986	1.766	[0.908, 3.479]	0.417
17	11	0.4301	0.4353	+1.2	2.325	2.114	[1.161, 3.887]	0.229
18	7	0.6604	0.4656	−29.5	1.514	1.298	[0.609, 2.825]	0.714
19	10	0.3161	0.2971	−6.0	3.164	2.847	[1.517, 5.405]	0.061
20	8	0.5300	0.5713	+7.8	1.887	1.651	[0.815, 3.402]	0.483

5.2 The pooled estimate

Under scheme A, formulas (7)–(8) give

$$\tilde{\kappa} = 1.9648, \qquad 95\%\ \text{CI} = [1.7205,\ 2.2444], \qquad p = 0.142. \tag{14}$$

The point estimate is 10% above e^{γ}, and it is precisely this excess that used to be interpreted as the trace of a factor omitted by the heuristic.

Estimator (12) for scheme B gives (the interval and p-value are obtained by inverting a simulation of scheme B: for a grid of values of κ, scheme B is simulated with the observed frontier placement, and one seeks those κ at which the observed statistic $(M - B)/S$ falls on the 0.025 and 0.975 quantiles of the simulated distribution; 20 000 replications per point)

$$\hat{\kappa}_{\text{B}} = \frac{M - B}{S} = 1.8381, \qquad 95\%\ \text{CI} = [1.613,\ 2.130], \qquad p = 0.639. \tag{15}$$

The excess falls from 10.3% to 3.2%, and the p-value rises from 0.14 to 0.64. We hold that scheme B is closer to the way the data were obtained, and therefore regard (15) as the main result and (14) as an upper bound: the latter corresponds to the assumption $m = 0$, that is, to stopping the search exactly at a find. The truth lies between them, and without publication of the L_b this interval cannot be narrowed. Both estimates are consistent with LPW; they differ in whether the data call for an explanation of a residual excess. We conclude that they do not. The overall picture is shown in Figure 1: the confidence interval of every base covers both e^{γ} and both pooled estimates.

5.3 Homogeneity, and its power

Let H_0 be the model with a single κ and H_1 the model with free κ_b. From (6), since $\sum_b \hat{\kappa}_b D_b/\ln b = M = \hat{\kappa} S$, the likelihood-ratio statistic reduces to

$$\Lambda = 2 \sum_b N_b \ln \frac{\hat{\kappa}_b}{\hat{\kappa}} \overset{H_0}{\sim} \chi^2_{B-1}. \tag{16}$$

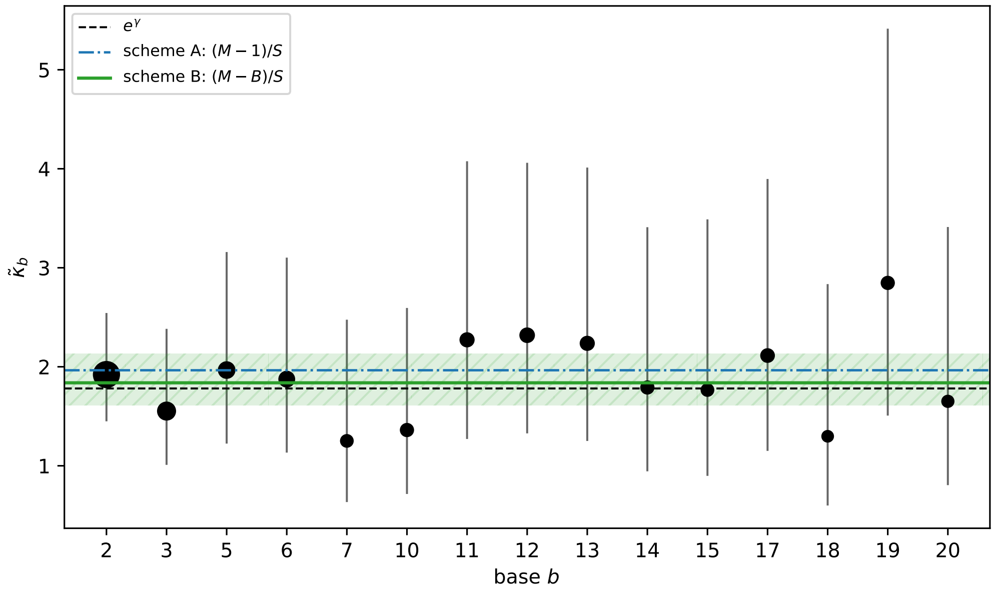

Figure 1. Per-base estimates $\tilde{\kappa}_b$ with exact 95% confidence intervals from Gamma$(N_b, 1)$; marker area is proportional to N_b. Dashed line: the LPW value e^γ; blue dash-dotted: the pooled estimate under scheme A; green with a hatched band: the estimate calibrated to scheme B and its interval. Each base separately is compatible with a universal constant.

We obtain $\Lambda = 7.40$ on 14 degrees of freedom, $p = 0.92$.

But a non-rejection is informative exactly to the extent that the test is powerful, and here one must apply to oneself the same requirement one applies to the literature. We simulated alternatives in which the κ_b are lognormally scattered about e^γ with coefficient of variation cv:

cv	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.50
power	0.061	0.120	0.212	0.363	0.535	0.693	0.817	0.893	0.965

The test attains 80% power only at $cv \approx 0.34$ (5000 replications per grid point, power accurate to about ± 0.01). Hence $p = 0.92$ means: the data do not require base-dependent frequencies, but neither do they exclude a spread of κ_b as large as a third of the value of the constant. The statement "lucky bases do not exist" does not follow from the available data; what follows is the weaker statement "their existence is not supported by the data, and differences below $\pm 34\%$ are in principle indistinguishable".

5.4 Robustness

Lower truncation. We varied the lower truncation point in order to probe the asymptotic regime (the heuristic (2) is asymptotic, whereas the constant-intensity model starts at $n = 2$). The results are in Table 6. The scheme A estimate grows from 1.965 to 2.099—not strictly monotonically (at $n > 10^2$ it retreats from 2.019 to 2.012) but systematically: the growth is 6.8%, and all five values come from nested subsamples of the same data. This is a trend, not noise. The scheme B estimate shows no trend and fluctuates about e^γ within 1.79–1.87 (under the weaker dropout criterion $N_b \geq 1$, within 1.76–1.87, still without a trend). The reason is visible from (12): truncation reduces M but not B, whereas scheme A subtracts one regardless of the sample size. We regard this as independent confirmation of the diagnosis of §4: the trend was a property of the estimator, not of the data. Both curves are shown in Figure 2.

Table 6. Robustness to lower truncation. A base is retained if at least *two* events survive the truncation ($N_b \geq 2$); at $n > 10^4$ base 19 no longer meets this criterion, whence $B = 14$. The criterion matters: under the weaker requirement $N_b \geq 1$ base 19 is retained and the last row becomes $B = 15$, $M = 90$, $(M-1)/S = 2.0848$, $(M-B)/S = 1.7569$, $p = 0.128$.

truncation	B	M	$(M-1)/S$	$(M-B)/S$	p (scheme A)
none	15	218	1.9648	1.8381	0.142
$n > 10$	15	192	2.0192	1.8712	0.080
$n > 10^2$	15	156	2.0116	1.8299	0.124
$n > 10^3$	15	125	2.0551	1.8231	0.106
$n > 10^4$	14	89	2.0990	1.7889	0.116

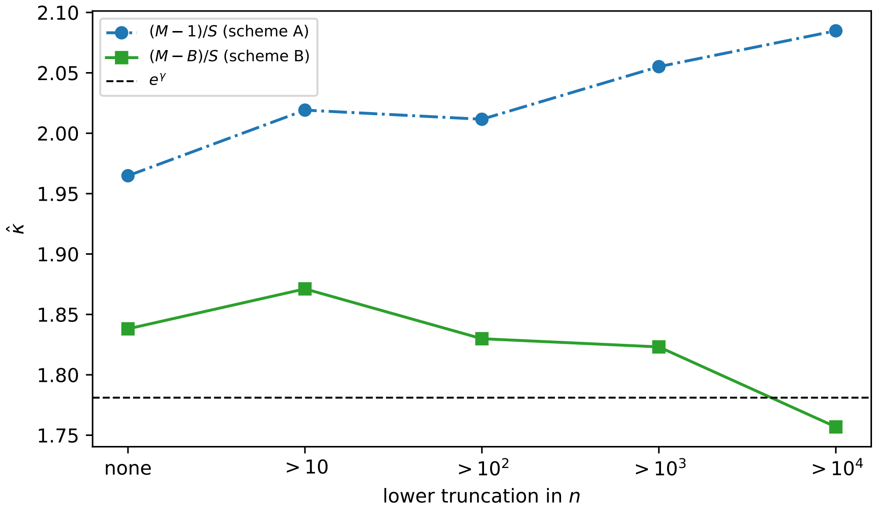

Figure 2. The pooled estimate under lower truncation: $(M-1)/S$ (scheme A) and $(M-B)/S$ (scheme B). The trend of the scheme A estimate disappears under scheme B.

Leave-one-base-out. A leave-one-base-out analysis gives $\tilde{\kappa}$ within $[1.930, 2.012]$ and p between 0.086 and 0.237; no single base determines the conclusion, and the most influential one, $b = 3$, shifts p only to 0.086 when removed.

Status of the data. Restricting base 2 to the fifty double-checked exponents moves the result from $\tilde{\kappa} = 1.965$, $p = 0.142$ to $\tilde{\kappa} = 1.961$, $p = 0.151$. Excluding all indices $n > 10^6$, where the entries have PRP status, gives $\tilde{\kappa} = 1.951$ with $M = 182$. In the worst case, when the last known PRP of some base turns out to be composite, the largest shift comes from $b = 17$: $\tilde{\kappa} = 1.977$. All variants lie inside interval (14), so possible pseudoprimes do not affect the conclusion.

6 Statistical power

By (8) the power against a true constant $\kappa = re^{\gamma}$ at level α is available in closed form (Table 7).

At 80% power the Mersenne sequence alone detects only deviations of 49.8% and above; pooling fifteen bases lowers that threshold to 21.2%. Attaining 80% power against a 10% deviation would require $M \approx 875$ events, and against 5%, $M \approx 3319$.

These thresholds are computationally out of reach, and it is worth saying by how much. Since M grows logarithmically in the search frontier, doubling the frontiers *of all fifteen bases* yields about 9.6 additional events for the entire pooled sample. To gather the missing 657 events one needs $657/9.6 \approx 68.2$ doublings, that is, a growth of the frontiers by a factor of $2^{68.2} \approx 3 \cdot 10^{20}$.

Table 7. Power at level $\alpha = 0.05$ for rejecting $\kappa = e^\gamma$ when the true constant exceeds it by the stated factor.

κ/e^γ	Mersenne only ($M = 51$)	pooled ($M = 218$)
1.05	0.060	0.106
1.10	0.095	0.277
1.15	0.152	0.523
1.20	0.229	0.755
1.30	0.427	0.972
1.50	0.802	1.000

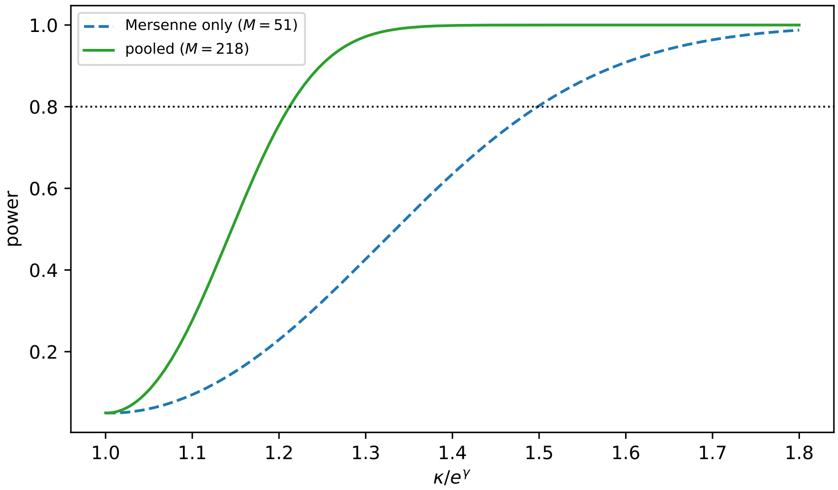

Figure 3. Power of rejecting $\kappa = e^\gamma$ at level $\alpha = 0.05$. The horizontal line marks 80% power.

This is not a matter of decades but a magnitude lying beyond the reach of any conceivable computation. Both power curves, as functions of the true deviation, are shown in Figure 3.

Hence a reading rule for this literature. A report that the data agree with LPW to within a few percent is not evidence for LPW at the level of a few percent; it is compatible with a constant that is wrong by a factor of one and a half. The pooled analysis does, however, establish a genuine bound: the universal constant, if it exists, lies roughly in [1.61, 2.13], and no base deviates from it detectably.

7 The same error in a different guise

The Mersenne counting function invites a related error, which we present as a worked example. Regressing $\pi_2(t)$ on t at the event points, over the 50 double-checked exponents (numbered $0, \ldots, N-1$: the smallest exponent is the origin, not an event), gives a slope of 2.6391 and an intercept of -2.9627 with an OLS standard error of 0.4469—apparently significant, and suggestive of a secondary term in (2). Nothing of the sort.

Two defects compound. The residuals of the cumulative count are strongly autocorrelated, here $\rho_1 = 0.825$; the Newey–West covariance estimator with eight lags raises the standard error to 0.7315. More importantly, the regression uses only the event points and thereby discards the empty interval between the last exponent and the search frontier. The correct null model holds the interval $[t_0, T]$ fixed, where $T = \ln X_{\mathrm{ver}}$, and conditions on exactly $N - 1 = 49$ events strictly to the right of t_0, which by Proposition 1 means 49 independent uniform points on $(t_0, T]$. Over 20 000 such replications the intercept has median -1.498 and a 5–95% range of $[-5.99, 2.20]$, giving $p = 0.562$ for the observed value. The secondary term is an artifact of conditioning.

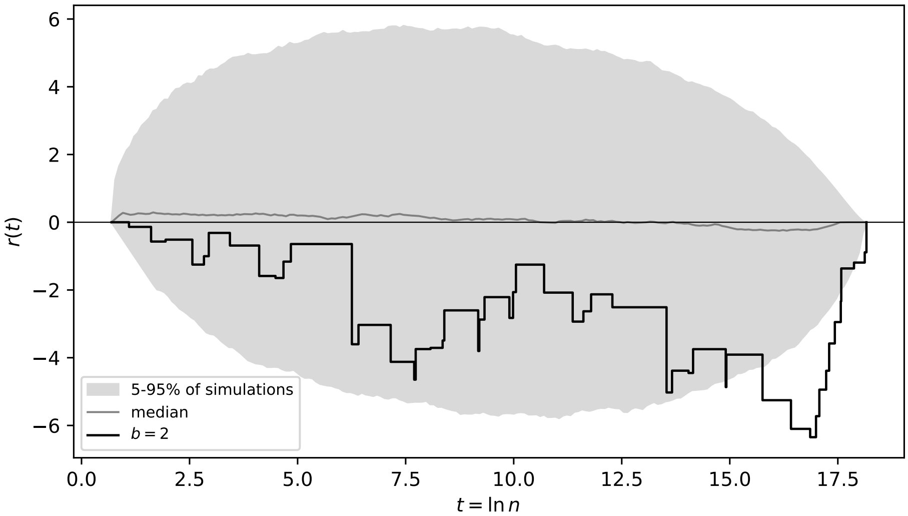

Figure 4. The residual process $r(t)$ for base $b = 2$ together with the 5–95% quantile band of 20 000 simulations and the simulation median. The trajectory stays inside the band.

7.1 Three consequences of one and the same conditioning

(i) The bootstrap does not help. The natural reaction to a suspiciously small OLS standard error is to bootstrap over the event points. On the base-2 data, a bootstrap over 20 000 resamples gives median -2.949, standard deviation 0.387 and a 95% interval $[-3.822, -2.300]$ lying entirely below zero; the proportion of negative values is 100%. This is not a confirmation but a repetition of the same error: resampling the observed points reproduces the variability within the realized configuration and cannot recover the variability of the process that produced that configuration. The correct null distribution is about six and a half times wider than the bootstrap one.

(ii) Spurious underdispersion. Comparing the observed $\max_i |r_i| = 6.347$, where $r_i = i - \hat{\lambda}(t_i - t_0)$, with a simulation of an *unconditioned* Poisson process suggests the conclusion that the counting function is "smoother than Poisson". Against the correct null distribution—conditioning on $N - 1$ events and estimating λ from the last of them—the median is 5.134 and the two-sided significance level is 0.539. The observation is entirely ordinary.

(iii) The "mid-range deficit". Estimating λ from the last event forces the residual r to vanish at both ends ($r(t_0) = r(t_N) = 0$ under the correct numbering), turning its trajectory into an analogue of a Brownian bridge, for which a negative excursion in the middle of the range arises of its own accord. The observed mean excursion over the middle half of the range is -3.13 against a simulation median of -0.001 and $p = 0.280$; the proportion of replications with a negative mean excursion is 0.500, so there is no systematic deficit. The trajectory of the residual process itself, together with the simulation band, is shown in Figure 4.

7.2 Base 2 at a known frontier

Base 2 is the only one whose frontier is published, and hence the only one for which a scheme B analysis is possible with no assumptions at all. With $T = \ln X_{\mathrm{ver}}$ we obtain $\hat{\lambda} = (N-1)/(T - t_0) = 2.796$ against $e^{\gamma}/\ln 2 = 2.570$, that is, $\hat{\kappa}_2 = 1.938$ with an exact 95% Garwood interval $[1.434, 2.562]$ and a Poisson p-value of 0.59 (doubled tail; under the mid-p convention, 0.54—the convention must be stated here, since the difference is appreciable). The conditional estimate on *the same slice of data*—the same 50 double-checked exponents—gives $\tilde{\kappa}_2 = 1.905$; the difference is

Table 8. Comparison of structural models. The likelihood is the conditional one (6) with factor C_b; LR and p are relative to model A; BIC is computed with $n = M = 218$.

Model	parameters	$\ln L$	LR	p	AIC	BIC	$P(\text{model} \mid \text{data})$
A	1	−176.129	—	—	354.26	357.64	0.893
B	3	−175.494	1.27	0.530	356.99	367.14	0.008
C	2	−175.647	0.96	0.326	355.29	362.06	0.098
D	4	−174.925	2.41	0.492	357.85	371.39	0.001

small because the last exponent of that slice, 77 232 917, happens to lie very close to the frontier 81 648 221. (For comparison, the conditional estimate over all 52 known exponents, reported in Table 5, is 1.921; it must not be compared with $\hat{\kappa}_2$, being a different sample.) This is a coincidence of one realization, not a property of the method: the bias of §4 concerns the average over the position of the last event relative to the frontier.

7.3 The theoretical secondary term has the opposite sign

There is an independent argument against a substantive reading of the intercept C_1. If the secondary term in (2) originated in the Mertens constant $\mathfrak{M} = 0.26150\ldots$, its natural size would be $(e^\gamma/\ln 2)\mathfrak{M} = +0.672$: the sign is positive, whereas what is observed is -2.963. The nearest theoretical candidates not merely fail to reproduce the observed magnitude—they predict the opposite sign. Independently of the foregoing, this confirms that C_1 is a parameter of the fit and not of the structure.

8 Structural corrections

Consider the more general parametrization of the intensity $\lambda_b = (\kappa/\ln b)\, C_b$ and four models for the factor:

$$\begin{aligned}
&\text{A:}\quad C_b = 1;\\
&\text{B:}\quad C_b = \prod_{p\mid b-1,\ p\neq 2} \left(\tfrac{p}{p-1}\right)^{\alpha} \prod_{p\mid b+1,\ p\neq 2} \left(\tfrac{p}{p-2}\right)^{\beta};\\
&\text{C:}\quad C_b = \prod_{p\mid b^2-1} \left(\tfrac{p}{p-1}\right)^{\delta};\\
&\text{D:}\quad C_b = \exp\big(\theta_1\omega(b-1) + \theta_2\omega(b+1) + \theta_3 \ln b\big).
\end{aligned}$$

The exponent in model C is written δ rather than γ so as not to confuse it with the Euler constant. The results are in Table 8. No parametrization gives a significant improvement; the BIC-based Bayes factors overwhelmingly support the base model.

Remark 5 (Sensitivity of the BIC). The choice of effective sample size for the BIC is not obvious: the likelihood (6) has $B = 15$ independent summands, but the information grows as $M = 218$. With $n = 15$ the posterior probabilities are 0.624 / 0.079 / 0.261 / 0.036; with $n = 218$ they are 0.893 / 0.008 / 0.098 / 0.001. The conclusion favors model A under either choice, but with $n = 218$ it is substantially stronger. We report $n = M$ as matching the number of independent events.

A fixed arithmetic correction. We considered separately a model with a *fixed* (not estimated) factor $C_b^{\text{fix}} = \prod_{p\mid b-1, p\neq 2} \frac{p}{p-1} \prod_{p\mid b+1, p\neq 2} \frac{p}{p-2}$, for which $\overline{C_b} = 2.46$. It gives $\hat{\kappa} = 0.815$. We stress that comparing this value with e^γ is meaningless: in the reparametrized model κ has a different scale, and such a "goodness-of-fit test" would only be testing whether the mean of C_b

equals one. Since models A and C^{fix} are non-nested and have the same number of free parameters, the correct comparison is directly by likelihood: $\ln L = -193.55$ against -176.13, a difference of -17.42. The fixed correction is decisively rejected.

Logarithmic corrections. The model $\lambda_b(t) = (\kappa/\ln b)(1 + c_1/t + c_2/t^2)$ is estimated by the full point-process likelihood $\sum_i \ln \lambda(t_i) - \int \lambda$ (the likelihood of the count alone is inadmissible here: it discards the positions of the events, which are the only thing that determines c_1, c_2). We obtain $\hat{\kappa} = 2.049$, $\hat{c}_1 = +0.108$, $\hat{c}_2 = -0.826$, LR $= 3.53$ on two degrees of freedom, $p = 0.171$. Not significant.

9 Conclusions

1. The estimators of the normalized frequency of generalized repunit primes that are in common use are *not* biased by the fitting scheme: both the endpoint form and the OLS slope are exactly unbiased. Only the reciprocal is biased, and that is a consequence of convexity. The common explanation of "slow convergence from below" in terms of censoring does not hold.

2. The genuine defect of current practice is that different forms of the estimator disagree by up to 29.5%, and that point values are published without confidence intervals even though the OLS standard errors in this problem are understated by a factor of about 1.6. The classification of bases into "lucky" and "unlucky" flips when the form of the estimator is changed on unchanged data.

3. Pooling $B = 15$ bases yields $M = 218$ events and the exact pivot $\kappa S \sim \text{Gamma}(M, 1)$, but that pivot is calibrated to the observation scheme "until the N-th event", whereas the data arise from a fixed search frontier. Under the actual scheme a nominal 5% test rejects in 16.2% of cases, and the correct pooled estimator is $(M - B)/S$: the conditional likelihood costs one event per base, not one event in total. Distinguishing such schemes goes back to reliability theory; what is new here is the case of an undocumented frontier, where the exposure cannot be taken from a protocol and has to be reconstructed from the finds (Remark 3).

4. This gives $\kappa = 1.838$ with a 95% interval $[1.61, 2.13]$ and $p = 0.639$ against e^γ. The reported ten-percent excess over LPW—including our own in an earlier version of this work—is an artifact of the observation scheme. Moving to scheme B independently removes the systematic trend of the pooled estimate under truncation in $n_{\min}$.

5. The choice between $(M-1)/S$ and $(M-B)/S$ is not settled by the data: both estimators are functions of the same sufficient statistics and differ by 7%. The practical consequence is the need to publish the search frontiers L_b, as GIMPS does for $b = 2$. Until that is done, any pooled analysis of these sequences carries an irreducible uncertainty of order 7%—twice the residual gap with LPW.

6. The LPW conjecture is not rejected by the available data, and base-dependent corrections are not required. But both of these are statements about the low sensitivity of the tests: the homogeneity test resolves only a spread of κ_b of order $\pm 34\%$, and the pooled test only deviations of the constant of at least 21%. Attaining a sensitivity of 10% would require the search frontiers to grow by a factor of about $3 \cdot 10^{20}$.

A Verification of the source data

All twenty sequences (fifteen of the main set and five from the check of the selection limit) were cross-checked against the OEIS entries on 27 August 2026. Since only the triples $(N_b, n_{\min}, n_{\max})$ enter the pooled analysis, Table 9 gives, for the main set, the number of known primes—from which N_b is obtained by subtracting one—and both extreme indices; the full lists of indices are contained in the script `analysis.py`.

Table 9. Cross-check against OEIS. The column "last addition" gives the provenance of the largest known index.

b	OEIS	primes	$n_{\min}$	$n_{\max}$	last addition
2	A000043	52	2	136 279 841	GIMPS, Oct. 2024
3	A028491	23	3	8 530 117	P. Bourdelais
5	A004061	19	3	3 300 593	P. Bourdelais
6	A004062	17	2	3 360 347	P. Bourdelais
7	A004063	10	5	1 264 699	P. Bourdelais
10	A004023	11	2	8 177 207	Propper and Batalov, May 2021
11	A005808	13	17	1 868 983	P. Bourdelais
12	A004064	14	2	769 543	P. Bourdelais, Dec. 2014
13	A016054	13	5	1 503 503	P. Bourdelais
14	A006032	11	3	1 724 417	P. Bourdelais, 13 Jul. 2026
15	A006033	10	3	639 833	P. Bourdelais, 22 Apr. 2019
17	A006034	12	3	1 990 523	P. Bourdelais, 3 Aug. 2020
18	A133857	8	2	1 270 141	P. Bourdelais
19	A006035	11	19	209 359	P. Bourdelais, 27 Aug. 2010
20	A127995	9	3	984 349	P. Bourdelais

A separate warning to the reader who checks the data against secondary sources: summary tables (the "Repunit primes" page in the OEIS Wiki, the article "List of repunit primes" in Wikipedia) are updated with a delay and, at the time of writing, lagged the OEIS entries themselves by one term for the bases $b = 14, 15, 17, 18, 20$. The OEIS entries should be treated as authoritative.

Independent recomputation of the prefixes

Cross-checking against OEIS confirms that the listed indices give primes, but it does not guarantee *completeness*: a missing term would shift N_b, and with it both pooled estimates. We closed this gap by direct recomputation. For each of the twenty bases $2 \le b \le 26$ that are not perfect powers, an independent implementation[2] tested *all* indices $n < 10^4$ consecutively, not only those listed in OEIS. Every negative verdict was recomputed by a second, independent backend (a blanket double-check, as in GIMPS) so as to rule out a lost find.

The result: 153 terms confirmed, none spurious, none missing, no disagreement between backends—across all twenty bases. Coverage is uneven across bases (from one term at $b = 18$ to twenty-two at $b = 2$): the main set $b \le 20$ accounts for 128 confirmed indices out of 233 known. Above 10^4 a complete recomputation is computationally out of reach, and there we still rely on OEIS.

We note that this recomputation was not a formality: on the first run the independent implementation declared $R_{3181}(23)$ composite, disagreeing with A204940. Analysis showed an error in the implementation itself—a spike of FFT rounding error on the last iteration, after the

[2] A hybrid CPU/GPU searcher: segmented sieving by small divisors, trial factoring in Montgomery arithmetic, PRP testing via GMP and GWNUM/IBDWT. The source code and the run logs are in the same repository, scripts `verify_sequences.sh` and `compare_verify.py`.

last periodic check—and not in the OEIS data. After the fix no disagreements remained. We report this episode as an illustration of the thesis of §4: a false "composite" is invisible without recomputation, and that is exactly why frontiers and verification protocols should be published.

Declarations

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Competing interests

The author declares no competing interests.

Data and code availability

All sequences used are listed in Table 1. The repository `https://github.com/Wilps93/repunit-hunt`, directory `paper/`, contains everything needed to reproduce the paper; the file tree and the order of execution are described in `paper/README.md`.

Data. The script `fetch_data.sh` downloads the OEIS b-files for all fifteen sequences of the main set, and `fetch_extra.sh` does the same for the five bases $21 \le b \le 26$ used in the check of the selection limit (§2.2). The downloaded files are included in the repository, so the analysis reproduces without network access.

Analysis. A single self-contained script `analysis.py` (numpy, scipy, matplotlib); every number in this paper is tagged in its output (`results.txt`) with the corresponding table or section and is reproduced by one run. The random number generator is initialized with a fixed seed, so all Monte Carlo quantities reproduce bit for bit. The script also produces all four figures. A full run takes a few minutes.

Independent recomputation of the sequences. `verify_sequences.sh` runs the searcher over all indices $n < 10^4$ for each base, and `compare_verify.py` compares the result with the OEIS b-files, returning a non-zero exit code on any disagreement (Appendix A).

A caveat on the Monte Carlo quantities. The endpoints of interval (15) depend on the inversion procedure, and Tables 2, 3 and 4 depend on the rules for placing the frontier. Both are described in the text and implemented in `analysis.py`; every value reported in this paper is exactly the output of that script under the seed fixed in it. Simulation accuracy: about ± 0.003 for the significance level in Table 4, and about $\pm 0.3\%$ for the means in Table 3.

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